

Learning goals: Installation of R, installing packages in R and loading libraries in R, introduction to R-studio, basics of R including arithmetic and logical operations, introduction to objects and functions

EXERCISES

- (1) **Arithmetic operations and operator precedence** Predict and check your results for the following arithmetic operations
 - (i) $2.7/2$
 - (ii) $2.7\% / \%2$ (What does $2.7 \% \% 2$ give you?)
 - (iii) $10+5i/2$
 - (iv) `round(2.5)`
 - (v) `round(-2.5)`
 - (vi) $2\% / \%4-1$
 - (vii) $3*2**2$ (equivalent to $3*2^2$)
 - (viii) $3**2*2$
 - (ix) $7\% / \% 4$
 - (x) $7\%\%4$
 - (xi) $-7\%\%4$
 - (xii) `trunc(5.7)`
 - (xiii) `trunc(-5.7)`
- (2) We know the result of `ceiling(5.7)`, which is 6. Write a one-line function to do the same. (Hint: You have to add 0.5 to the number and use the `floor` function)
- (3) **Logical operations** Predict and check your results for the following boolean operations. Assign the values 1,2,4 respectively to a,b and c and do the following
 - (i) `a and b`
 - (ii) `not a<b or c>b`
- (4) **Intricacies of the integer function** This exercise is to teach you to not change the data type of a series of numbers using the `integer` function. What is the result of
 - (i) `x<- c(5,3,7,8)` followed by
 - (ii) `is.integer(x)` followed by
 - (iii) `is.numeric(x)`
 - (iv) Try `x<- integer(length(x))` and print `x`. What has happened?
 - (v) Repeat the above except the last step and do `x<-as.integer(x)` followed by `is.integer(x)`. What is the result? Why did you get this result?
- (5) **Testing equality of two real numbers** This exercise is to teach you to NOT USE the `==` operator for comparing two real numbers.
 - (i) `x <- sqrt(2)`
 - (ii) `x*x ==2` This is expected to be TRUE, but its not always! Why? What's the difference? So try
 - (iii) `x*x - 2` gives you the difference. This is the error due to rounded off representation of real numbers.
 - (iv) `all.equal(x*x,2)` gives you the correct answer.